Deep Learning (Spring 2026)
Solution for Assignment 7

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Solution 1. True.
Adding noise to the continuous embedding vectors of a sequence of discrete tokens (words) and then applying a reverse diffusion process (with conditional resampling/rounding to map back to discrete tokens) is a valid approach for text generation using diffusion models (e.g., as proposed in Diffusion-LM).

Solution 2.

1. **Prove $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$:**
   We have the forward process step:

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_{t-1}$$

   Substituting $x_{t-1} = \sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-2}$ into the equation for x_t:

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\left(\sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-2}\right) + \sqrt{1-\alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \left(\sqrt{\alpha_t(1-\alpha_{t-1})}\epsilon_{t-2} + \sqrt{1-\alpha_t}\epsilon_{t-1}\right) \end{aligned}$$

   Since ϵ_{t-2} and ϵ_{t-1} are independent standard normal variables, the linear combination of them is also Gaussian with mean 0 and variance:

$$\alpha_t(1-\alpha_{t-1}) + (1-\alpha_t) = \alpha_t - \alpha_t\alpha_{t-1} + 1 - \alpha_t = 1 - \alpha_t\alpha_{t-1}$$

   Thus, $x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\tilde{\epsilon}_{t-2}$, where $\tilde{\epsilon}_{t-2} \sim \mathcal{N}(0, I)$. By repeating this substitution process recursively down to x_0, we obtain:

$$x_t = \sqrt{\prod_{i=1}^{t}\alpha_i}\,x_0 + \sqrt{1-\prod_{i=1}^{t}\alpha_i}\,\epsilon = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

   This shows that $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$.

2. **Prove $q(x_{t-1}|x_t, x_0)$ is a Gaussian with mean $\tilde{\mu}_t$:**
   Using Bayes' theorem and the Markov property:

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

   Since all distributions on the right-hand side are Gaussian, $q(x_{t-1}|x_t, x_0)$ is also Gaussian. We can find its mean and variance by analyzing the exponent of the product of these Gaussian PDFs. The exponent (ignoring constants) is:

$$-\frac{1}{2}\left(\frac{\|x_t-\sqrt{\alpha_t}x_{t-1}\|^2}{1-\alpha_t}+\frac{\|x_{t-1}-\sqrt{\bar{\alpha}_{t-1}}x_0\|^2}{1-\bar{\alpha}_{t-1}}-\frac{\|x_t-\sqrt{\bar{\alpha}_t}x_0\|^2}{1-\bar{\alpha}_t}\right)$$
$$=-\frac{1}{2}\left(x_{t-1}^T\left[\frac{\alpha_t}{1-\alpha_t}+\frac{1}{1-\bar{\alpha}_{t-1}}\right]Ix_{t-1}-2x_{t-1}^T\left[\frac{\sqrt{\alpha_t}}{1-\alpha_t}x_t+\frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0\right]+C(x_t,x_0)\right)$$

The precision (inverse variance) matrix is $\left(\frac{\alpha_t(1-\bar{\alpha}_{t-1})+(1-\alpha_t)}{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}\right)I=\frac{1-\bar{\alpha}_t}{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}I$. Thus, the variance is $\tilde{\beta}_t=\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$. The mean $\tilde{\mu}_t$ is obtained by multiplying the variance with the linear coefficient:

$$\tilde{\mu}_t=\tilde{\beta}_t\left(\frac{\sqrt{\alpha_t}}{1-\alpha_t}x_t+\frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0\right)=\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t+\frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}x_0$$

From the reparameterization in part 1, we know $x_0=\frac{x_t-\sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\bar{\alpha}_t}}$. Substituting x_0 into the mean formula and using $\sqrt{\bar{\alpha}_t}=\sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}$:

$$\tilde{\mu}_t=\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t+\frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\left(\frac{x_t-\sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}}\right)$$
$$=\left(\frac{\alpha_t(1-\bar{\alpha}_{t-1})+1-\alpha_t}{\sqrt{\alpha_t}(1-\bar{\alpha}_t)}\right)x_t-\frac{1-\alpha_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}\epsilon$$
$$=\frac{1}{\sqrt{\alpha_t}}x_t-\frac{1-\alpha_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}\epsilon=\frac{1}{\sqrt{\alpha_t}}\left(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right)$$

3. **Show the VLB inequality and equality:**
First, we derive the variational lower bound using the non-negativity of KL divergence:

$$-\log p_\theta(x_0)\le-\log p_\theta(x_0)+D_{\mathrm{KL}}(q(x_{1:T}|x_0)\|p_\theta(x_{1:T}|x_0))$$
$$=-\log p_\theta(x_0)+\mathbb{E}_{q(x_{1:T}|x_0)}\left[\log\frac{q(x_{1:T}|x_0)}{p_\theta(x_{1:T}|x_0)}\right]$$
$$=\mathbb{E}_{q(x_{1:T}|x_0)}\left[\log\frac{q(x_{1:T}|x_0)}{p_\theta(x_{1:T}|x_0)p_\theta(x_0)}\right]=\mathbb{E}_{q(x_{1:T}|x_0)}\left[\log\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}\right]$$

Taking the expectation $\mathbb{E}_{q(x_0)}$ on both sides gives the first inequality:

$$\mathbb{E}_{q(x_{0:T})}\left[\log\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}\right]\ge-\mathbb{E}_{q(x_0)}\log p_\theta(x_0)$$

Next, we expand the term inside the expectation:

$$\log\frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})}=\log\frac{\prod_{t=1}^T q(x_t|x_{t-1})}{p_\theta(x_T)\prod_{t=1}^T p_\theta(x_{t-1}|x_t)}$$

Using Bayes' rule $q(x_t|x_{t-1})=q(x_t|x_{t-1},x_0)=\frac{q(x_{t-1}|x_t,x_0)q(x_t|x_0)}{q(x_{t-1}|x_0)}$ for $t\ge 2$, the product of forward conditionals can be rewritten as a telescoping product:

$$\prod_{t=1}^T q(x_t|x_{t-1})=q(x_1|x_0)\prod_{t=2}^T\frac{q(x_{t-1}|x_t,x_0)q(x_t|x_0)}{q(x_{t-1}|x_0)}=q(x_T|x_0)\prod_{t=2}^T q(x_{t-1}|x_t,x_0)$$

Substituting this back into the log ratio:

$$\log \frac{q(x_{1:T}|x_0)}{p_\theta(x_{0:T})} = \log \left(\frac{q(x_T|x_0)}{p_\theta(x_T)} \cdot \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \cdot \frac{1}{p_\theta(x_0|x_1)} \right)$$
$$= \log \frac{q(x_T|x_0)}{p_\theta(x_T)} + \sum_{t=2}^{T} \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} - \log p_\theta(x_0|x_1)$$

Taking the expectation over $q(x_{0:T})$, the first term becomes $D_{\mathrm{KL}}(q(x_T|x_0)\|p_\theta(x_T))$ which is L_T. The summation term becomes $\sum_{t=2}^{T} D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)\|p_\theta(x_{t-1}|x_t))$ which are the L_{t-1} terms, and the last term becomes $-\log p_\theta(x_0|x_1)$ which is L_0. This completes the proof.

4. **Prove the rewritten formula for L_t:**
   We are given:

$$L_t = \mathbb{E}_{x_0,\epsilon} \left[\frac{1}{2\|\Sigma_\theta\|_2^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right]$$

   Substituting the expressions for $\tilde{\mu}_t$ (from part 2) and $\mu_\theta(x_t, t)$:

$$\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right) - \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$
$$= -\frac{1-\alpha_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \big(\epsilon - \epsilon_\theta(x_t, t)\big)$$

   Taking the squared L_2 norm of this difference:

$$\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 = \frac{(1-\alpha_t)^2}{\alpha_t(1-\bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2$$

   Substituting this back into L_t and noting that $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$:

$$L_t = \mathbb{E}_{x_0,\epsilon} \left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\|\Sigma_\theta\|_2^2} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t) \right\|^2 \right]$$

   which is exactly the target formula.

Solution 3. Expanding the squared L_2 norm inside the expectation of the Fisher divergence:

$$F(p_{\text{data}}\|p_\theta) = \frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}} \left[\|\nabla_x \log p_{\text{data}}(x) - \nabla_x \log p_\theta(x)\|_2^2 \right]$$
$$= \frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}} \left[\|\nabla_x \log p_{\text{data}}(x)\|_2^2 + \|\nabla_x \log p_\theta(x)\|_2^2 - 2\nabla_x \log p_{\text{data}}(x)^T \nabla_x \log p_\theta(x) \right]$$

The first term $\frac{1}{2}\mathbb{E}_{x\sim p_{\text{data}}}\|\nabla_x \log p_{\text{data}}(x)\|_2^2$ does not depend on θ and can be absorbed into the constant *Const.*. Now, consider the cross term:

$$-\mathbb{E}_{x\sim p_{\text{data}}} \left[\nabla_x \log p_{\text{data}}(x)^T \nabla_x \log p_\theta(x) \right] = -\int p_{\text{data}}(x) \left(\frac{\nabla_x p_{\text{data}}(x)}{p_{\text{data}}(x)} \right)^T \nabla_x \log p_\theta(x)\, dx$$
$$= -\int \nabla_x p_{\text{data}}(x)^T \nabla_x \log p_\theta(x)\, dx$$

We apply integration by parts to each dimension i:

$$\int \frac{\partial p_{\text{data}}(x)}{\partial x_i} \frac{\partial \log p_\theta(x)}{\partial x_i}\, dx = \left[p_{\text{data}}(x) \frac{\partial \log p_\theta(x)}{\partial x_i} \right]_{-\infty}^{\infty} - \int p_{\text{data}}(x) \frac{\partial^2 \log p_\theta(x)}{\partial x_i^2}\, dx$$

Under mild regularity conditions, $p_{\text{data}}(x) \to 0$ as $\|x\| \to \infty$, making the boundary term vanish. Thus:

$$
\begin{aligned}
-\int \nabla_x p_{\text{data}}(x)^T \nabla_x \log p_\theta(x)\, dx &= \int p_{\text{data}}(x) \sum_i \frac{\partial^2 \log p_\theta(x)}{\partial x_i^2}\, dx \\
&= \mathbb{E}_{x\sim p_{\text{data}}}\left[\operatorname{tr}(\nabla_x^2 \log p_\theta(x))\right]
\end{aligned}
$$

Putting it all together, we obtain:

$$
F(p_{\text{data}}\|p_\theta) = \mathbb{E}_{p_{\text{data}}}\left[\frac{1}{2}\|\nabla_x \log p_\theta(x)\|_2^2 + \operatorname{tr}(\nabla_x^2 \log p_\theta(x))\right] + Const.
$$

Solution 4. We start with the objective in denoising score matching:

$$
\int q_\sigma(\tilde{x}) \nabla_{\tilde{x}} \log q_\sigma(\tilde{x})^T s_\theta(\tilde{x})\, d\tilde{x}
$$

Using the identity $q_\sigma(\tilde{x})\nabla_{\tilde{x}} \log q_\sigma(\tilde{x}) = \nabla_{\tilde{x}} q_\sigma(\tilde{x})$, the objective becomes:

$$
\int \nabla_{\tilde{x}} q_\sigma(\tilde{x})^T s_\theta(\tilde{x})\, d\tilde{x}
$$

Since the perturbed data distribution $q_\sigma(\tilde{x})$ is obtained by convolving the data distribution $p_{\text{data}}(x)$ with the noise kernel $q_\sigma(\tilde{x}|x)$, we have $q_\sigma(\tilde{x}) = \int p_{\text{data}}(x) q_\sigma(\tilde{x}|x)\, dx$. Substituting this into the gradient:

$$
\nabla_{\tilde{x}} q_\sigma(\tilde{x}) = \nabla_{\tilde{x}} \int p_{\text{data}}(x) q_\sigma(\tilde{x}|x)\, dx = \int p_{\text{data}}(x) \nabla_{\tilde{x}} q_\sigma(\tilde{x}|x)\, dx
$$

Now substitute this back into the integral:

$$
\int \left(\int p_{\text{data}}(x) \nabla_{\tilde{x}} q_\sigma(\tilde{x}|x)\, dx \right)^T s_\theta(\tilde{x})\, d\tilde{x}
$$

By Fubini's theorem, we can swap the order of integration:

$$
\int p_{\text{data}}(x) \left(\int \nabla_{\tilde{x}} q_\sigma(\tilde{x}|x)^T s_\theta(\tilde{x})\, d\tilde{x} \right) dx
$$

Using the same log-derivative trick $\nabla_{\tilde{x}} q_\sigma(\tilde{x}|x) = q_\sigma(\tilde{x}|x)\nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x)$, we obtain:

$$
\begin{aligned}
\int p_{\text{data}}(x) \left(\int q_\sigma(\tilde{x}|x) \nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x)^T s_\theta(\tilde{x})\, d\tilde{x} \right) dx &= \mathbb{E}_{x\sim p_{\text{data}}(x)} \mathbb{E}_{\tilde{x}\sim q_\sigma(\tilde{x}|x)} \left[\nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x)^T s_\theta(\tilde{x})\right] \\
&= \mathbb{E}_{x\sim p_{\text{data}}(x), \tilde{x}\sim q_\sigma(\tilde{x}|x)} \left[\nabla_{\tilde{x}} \log q_\sigma(\tilde{x}|x)^T s_\theta(\tilde{x})\right]
\end{aligned}
$$

This completes the proof.

Solution 5. In DDPM, the forward diffusion process adds Gaussian noise progressively such that the distribution of the noisy state x_t given x_0 is:

$$
q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t)I)
$$

Equivalently,

$$
x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I).
$$

The conditional score of this Gaussian distribution is:

$$\nabla_{x_t} \log q(x_t|x_0) = \nabla_{x_t} \left(-\frac{\|x_t - \sqrt{\bar{\alpha}_t} x_0\|^2}{2(1-\bar{\alpha}_t)} + C \right) = -\frac{x_t - \sqrt{\bar{\alpha}_t} x_0}{1-\bar{\alpha}_t} = -\frac{\epsilon}{\sqrt{1-\bar{\alpha}_t}}$$

Thus, predicting the noise ϵ is equivalent to predicting the conditional score up to the known scalar factor $-\sqrt{1-\bar{\alpha}_t}$.

The usual DDPM training objective is:

$$\mathbb{E}_{x_0,\epsilon,t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right].$$

For a fixed x_t and t, the MSE-optimal prediction is $\epsilon_\theta(x_t, t) = \mathbb{E}[\epsilon \mid x_t]$. By the denoising score matching identity, the marginal score of the noisy distribution $q_t(x_t)$ satisfies:

$$\nabla_{x_t} \log q_t(x_t) = -\frac{\mathbb{E}[\epsilon \mid x_t]}{\sqrt{1-\bar{\alpha}_t}}.$$

Therefore the noise prediction network also defines a score estimator:

$$s_\theta(x_t, t) \approx \nabla_{x_t} \log q_t(x_t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1-\bar{\alpha}_t}}.$$

This is exactly the connection to Noise-Conditioned Score Networks (NCSN). In NCSN, one explicitly trains a score network $s_\theta(x, \sigma)$ for data distributions perturbed by a schedule of Gaussian noise levels $\sigma_1 > \sigma_2 > \cdots > \sigma_T$. DDPM also provides a sequence of noise levels through t, with effective noise variance controlled by $1 - \bar{\alpha}_t$. The difference is mostly parameterization: NCSN directly predicts the score, while DDPM usually predicts the added noise and converts it to a score by a known scaling factor. Both methods learn score fields over progressively less noisy data distributions and generate samples by starting from high noise and moving through the learned reverse denoising process toward the data distribution.

Solution 6. Used Codex (AI assistant by OpenAI) in the following capacities:

- **Solution ideation:** Brainstorming approaches and generating initial ideas for mathematical proofs, particularly for the DDPM variational lower bound derivation and the connection between DDPM and NCSN.
- **English grammar and writing:** Polishing and refining the English writing throughout the solutions to improve clarity, coherence, and academic tone.
- **LaTeX formatting:** Optimizing typesetting and layout of mathematical expressions and document structure for better readability and presentation.
- **Result verification:** Cross-checking the correctness of intermediate derivations, such as the Fisher Divergence integration by parts and the Denoising Score Matching expectation swap.

All final solutions were reviewed and verified by the author.